Benjamin Lobos Lertpunyaroj

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Education

Purdue University — **West Lafayette, IN**
Bachelor of Science in Computer Science — May 2027
Concentrations: Software Engineering, Systems Programming — GPA: 3.85/4.0
Minor: Business Management
- College of Science Deans List & Semester Honors: *2023, 2024, 2025*
- Relevant Coursework: *Data Structures and Algorithms (C++), OOP, Discrete Mathematics, Calculus, Linear Algebra, Software Engineering, Computer Architecture (Asm Arm64), Programming in C, Systems Programming.*

Skills

Languages C, C++, C#.Net, Python, JavaScript (TS, React), HTML/CSS (TailwindCSS), Java, Swift, Lua, VimScript, R, Bash, AppleScript, Assembly (x86, Arm64)
Tools LaTeX, Vi/Vim/Neovim, Zsh, Unity, Airflow, Linux, Debian, PostgreSQL, AArch64, Ubuntu, Unix, Git

Experience

C Programming Teaching Assistant @ Purdue University — **November 2024 – Present**
- Provided academic support to 750+ students (CS 24000), leading lab sessions, holding weekly office hours, and clarifying concepts related to the C programming language
- Developed a homework assignment from scratch to reinforce data structure fundamentals using LaTeX.
- Implemented multiple midterm and final practice exams into the class curriculum to enhance students' exam readiness and practical skills.

Software Engineering Intern @ Qservus — **December 2022 – March 2023**
- Conducted implementation of Qservus Lite for two months, a product oriented in quality of service, using AngularJS.
- Streamlined the deployment of the international version of Qservus' main website.
- Developed a mobile version (iOS) of a customer survey application using Swift.

Projects

SlayterHIL — C (protobuf-c), C++ (Physics engine, UDP), Raspberry Pi, RTOS — **August 2025 - Present**
Embedded Hardware-in-the-Loop Drone (UAV) Testbed
- Engineered a reproducible drone HIL platform to test off-aircraft flight software through simulating dynamics and automating runs/results through a Raspberry Pi-based orchestrator.
- Developed a C++ flight-physics engine for an RTOS, injecting deterministic telemetry and aggregating results over SPI using Protocol Buffers for controlled iterative aircraft testing.

Elite Edge — JavaScript (React), TailwindCSS, Python, PostgreSQL, Apache Airflow — **May - September 2025**
College Basketball Weekly Insights & Analytics
- Architected a Docker-containerized backend that scrapes weekly KenPom data (2013-25), stores it in PostgreSQL, and feeds Python scikit-learn models producing NCAA analytics with Apache Airflow for data workflow automation.
- Shipped mobile-responsive site with interactive excellence and similarity analytics using React and TailwindCSS.

CHIP-8 Emulator/interpreter — C (SDL2), C++ — **April – May 2025**
Computer Architecture Emulation
- Built a cycle-accurate virtual machine that implements the entire CHIP-8 instruction set, RAM, and program porting in C.
- Leveraged SDL2 to deliver 60 Hz graphics and sound, sustaining >2000 cycles/sec on x86_64 and ARMv8, in C++.

macOS Neovim — Bash, AppleScript — **February – April 2025**
Desktop Application
- Developed an app enabling users to open files directly in a Neovim instance or the configured default file explorer, automatically handling shell sessions and directory management, using Bash and AppleScript.
- Implemented customizable application builds supporting multiple terminal emulators (Terminal.app, iTerm), including automated naming, icon integration, and streamlined macOS code signing.

Interm.nvim & Mintex.nvim — Lua, VimScript, LaTeX — **August 2024 – January 2025**
Neovim Plugins
- Developed interm.nvim, a plugin that enhances terminal window management with custom status lines, Bash shell highlights, and optimized directory synchronization and shell handling, using Lua and Vimscript.
- Maintained mintex.nvim, a lightweight minimal LaTeX vim compiler to open a quick view PDF of the currently edited file

Publications

Unity: Front to Back — **May – September 2022**
Author of Published Book
- Published a technical book on C# and Unity game development (UI, Physics, Scripting, Events, Animation).
- Focused on detailed real development scenarios over verbosity for effective learning.
- Sold physical and digital copies on Amazon and other bookstores at affordable margins.